

UNIT – II

Cellular Radio System Design

2.1 GENERAL DESCRIPTION OF THE PROBLEM

2.2 CONCEPT OF FREQUENCY REUSE CHANNELS

A radio channel consists of a pair of frequencies one for each direction of transmission that is used for full-duplex operation. A Particular radio channel, say F_1 , used in one geographic zone to call a cell, says C_1 , with a coverage radius R can be used in another cell with the same coverage radius at a distance D away.

Frequency reuse is the core concept of the cellular mobile radio system. In this frequency reuse system, users in different geographic locations (different cells) may simultaneously use the same frequency channel (see Fig.2.2). The frequency reuse system can drastically increase the spectrum efficiency, but if the system is not properly designed, serious interference may occur. Interference due to the common use of the same channel is called *cochannel interference* and is our major concern in the concept of frequency reuse.

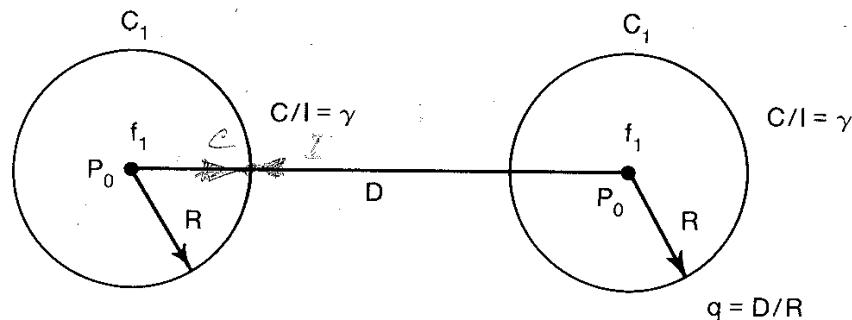


Figure 2.2 The ratio of D/R .

FREQUENCY REUSE SCHEME: The frequency reuse concept can be used in the time domain and the space domain. Frequency reuse in the time domain results in the occupation of the same frequency in different time slots. It is called time division multiplexing (TDM). Frequency reuse in the space domain can be divided into two categories.

- 1 Same frequency assigned in two different geographic areas, such as A.M or FM radio stations using the same frequency in different cities.

2 Same frequency repeatedly used in a same general area in one system - the scheme is used in cellular systems. There are many co-channel cells in the system. The total frequency spectrum allocation is divided into K frequency reuse patterns, as illustrated in Fig 2.3 for K = 4, 7, 12, and 19.

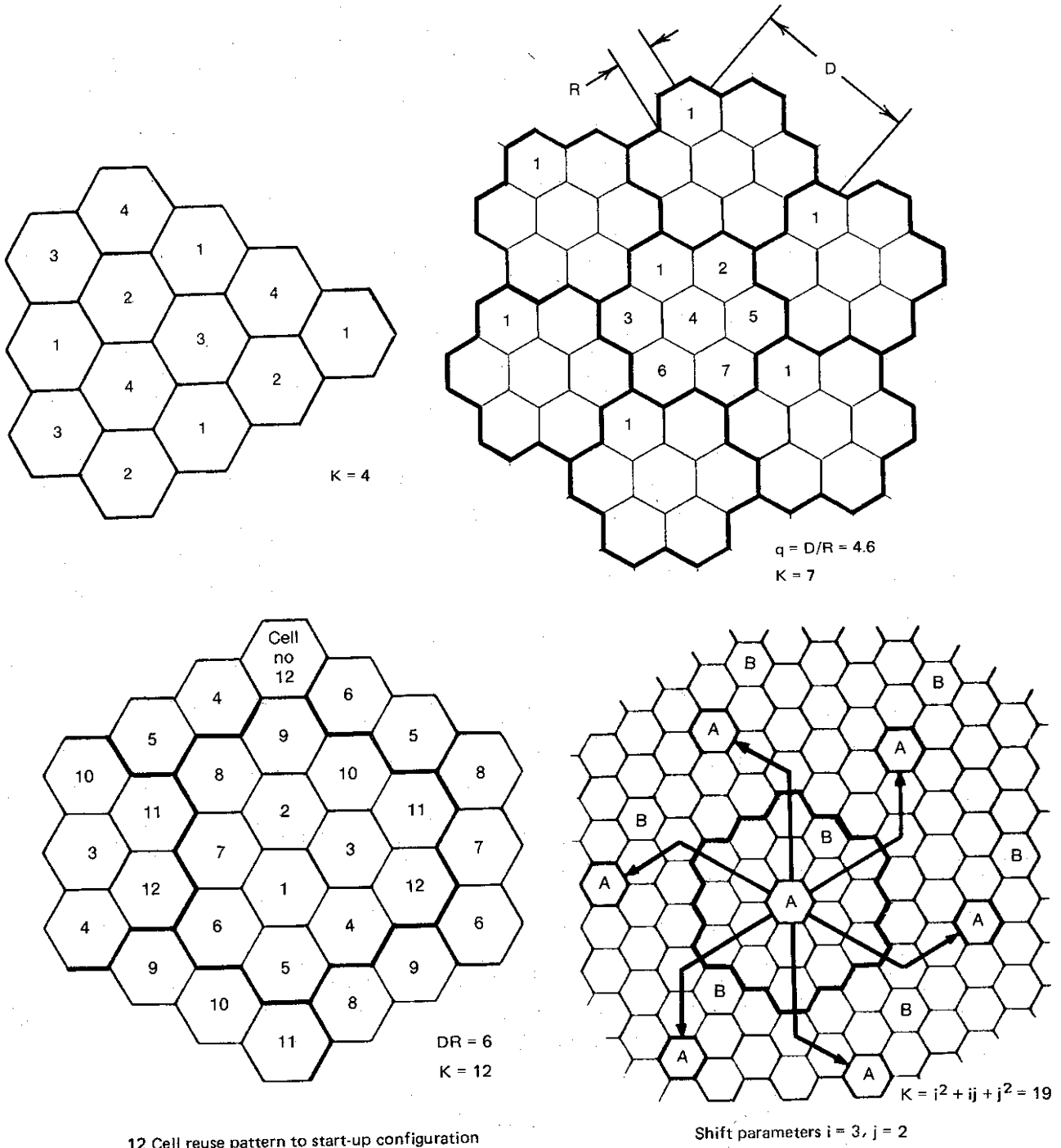


Figure 2.3 N-cell reuse pattern.

Frequency Reuse Distance

The minimum distance which allows the same frequency to be reused will

depend on the number of co-channel cells in the vicinity of the center cell, the type of geographic terrain contour, the antenna height, and transmitted power at each cell site. The frequency reuse distance D can be express by

$$D = \sqrt{3K} R$$

Where K is the frequency reuse pattern or cluster size shown in figure 2.3 and R is the radius of cell site.

$$D = \begin{cases} 3.46R & K = 4 \\ 4.6R & K = 7 \\ 6R & K = 12 \\ 7.55R & K = 19 \end{cases}$$

If all the cell sites transmit the same power, then K increases and the frequency reuse distance D increases. This increased D reduces the chance that cochannel interference may occur.

Theoretically, a large K is desired. However, the total number of allocated channels is fixed. When K is too large, the number of channels assigned to each of K cells becomes small. It is always true that if the total number of channels in K cells is divided as K increases, trunking inefficiency results. The same principle applies to spectrum inefficiency: if the total numbers of channels are divided into two network systems serving in the same area, spectrum inefficiency increases.

Obtaining the smallest number K involves estimating cochannel interference and selecting the minimum frequency reuse distance D to reduce cochannel interference. The smallest value of K is $K = 3$, obtained by setting $i = 1, j = 1$ in the equation $K = i^2 + i j + j^2$.

2.3 CO CHANNEL INTERFERENCE REDUCTION FACTOR

Frequency reuse technique is very useful for increasing the efficiency of spectrum utilization. But which results the cochannel interference since the same frequency channel is used repeatedly in different co channel cells. To reduce the co channel interference, we need to find the minimum frequency reuse distance. Assume that the all the cell sites are identical in size. The cell size is area where particular signal

strength is available in cell site. As the cell size is fixed then the cochannel interference is now independent of the transmitted power of each cell. It means that the received threshold level at the mobile unit is adjusted to the size of the cell.

The co channel interference is a function of a parameter q which is known as cochannel inference reduction factor and defined as

$$q = \frac{D}{R} \quad \text{.....1}$$

Where D is the frequency reuse distance and R is the radius of cell site. If value of q increases then co channel interference decreases.

Further on writing the D in Eq. 1 as a function of K_I and C/I we have

$$D = f (K_I, C/I) \quad \text{.....2}$$

Where K_I is the number of co channel interfering cells in the first tier, C is carrier signal power, I is the interference signal power and C/I is the received carrier-to-interference ratio at the desired mobile receiver.

$$\frac{C}{I} = \frac{C}{\sum_{k=1}^{K_I} I_k} \quad \text{...3}$$

In perfect hexagonal-shaped cellular system, in the first tier there are always six cochannel interfering cells, as shown in figure 2.4. In first tier the value of K_I is 6. The co channel interference will be experienced at both cell site and at mobile units in the center cell. If the interference is much greater, then the carrier-to-interference ratio at the mobile units caused by the six interfering sites is the same as the C/I received at the center cell site caused by interfering mobile units in the six cells. According to the reciprocity theorem and the statistical summation of radio propagation, the two C/I values can be very close. Assume that the local noise is much less than the interference level and can be neglected. The propagation path-loss slope variation in mobile radio environment is given by

$$C = \alpha R^{-\gamma} \quad \text{.....4}$$

C/I then can be expressed, on the basis of the propagation path-loss slope

$$\frac{C}{I} = \frac{R^{-\gamma}}{\sum_{k=1}^{K_I} D_k^{-\gamma}} \quad \text{.....5}$$

Where γ is a propagation path-loss slope determined by the actual terrain environment. In a mobile radio medium γ usually is assumed to be 4. K_I is the number of co channel interfering cells and is equal to 6 in a fully developed system, as shown in figure 2.4. The six co channel interfering cells in the second tier cause weaker interference than those in the first tier. Therefore, the cochannel interference from the second tier of interfering cells is negligible. Substituting Eq. 1 into Eq. 5 we have

$$\frac{C}{I} = \frac{1}{\sum_{k=1}^{K_I} \left(\frac{D_k}{R}\right)^{-\gamma}} = \frac{1}{\sum_{k=1}^{K_I} (q_k)^{-\gamma}} \quad \text{.....6}$$

Where q_k is the co channel interference reduction factor with k^{th} co channel interfering cell

$$q_k = \frac{D_k}{R}$$

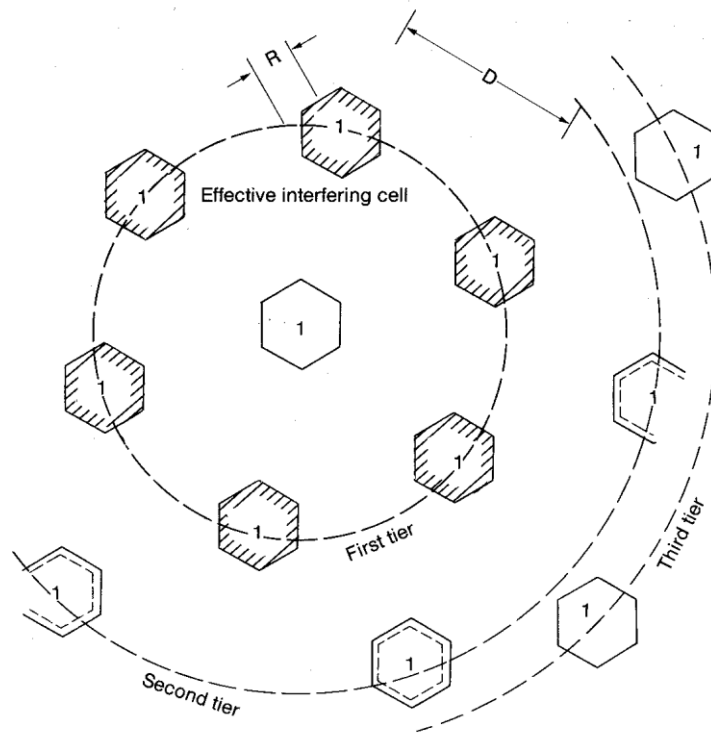


Figure 2.4 Six effective interfering cells of cell 1.

1.3 DESIRED C/I RATIO

In case of calculating the C/I for omnidirectional antenna system we considered following cases:

1. The co channel interference and signal received by mobile unit
2. The co channel interference and signal received by cell site.

Both cases 1 & 2 are shown in figure 2.5. The local noises at the cell site and the mobile unit is denoted by N_b and N_m respectively.

Normally the local noise is small as compared with the interference level so we can neglect N_m and N_b . If the received carrier-to-interference ratios at both the mobile unit and the cell site are the same, the system is called a balanced system. In a balanced system, we can choose either one of the two cases to analyze the system requirement; the results from one case are the same for the others. Assume that all D_k are the same for simplicity then $D = D_k$, and $q = q_k$,

$$\frac{C}{I} = \frac{R^{-\gamma}}{6D^{-\gamma}} = \frac{q^\gamma}{6} \quad \dots\dots 1$$

Thus

$$q^\gamma = 6 \frac{C}{I} \quad \dots\dots 2$$

$$q = \left(6 \frac{C}{I}\right)^{1/\gamma}$$

.....3

In equation 3 the value of C/I is based on the required system performance and the specified value of γ is based on the terrain environment. With given values of C/I and γ , the co channel interference reduction factor q can be determined. Normal cellular practice is to specify C/I to be 18 dB or higher based on subjective tests and the criterion described. Because a C/I of 18 dB is measured by the acceptance of voice quality from present cellular mobile receivers, this acceptance implies that both mobile radio multipath fading and co channel interference become ineffective at that level.

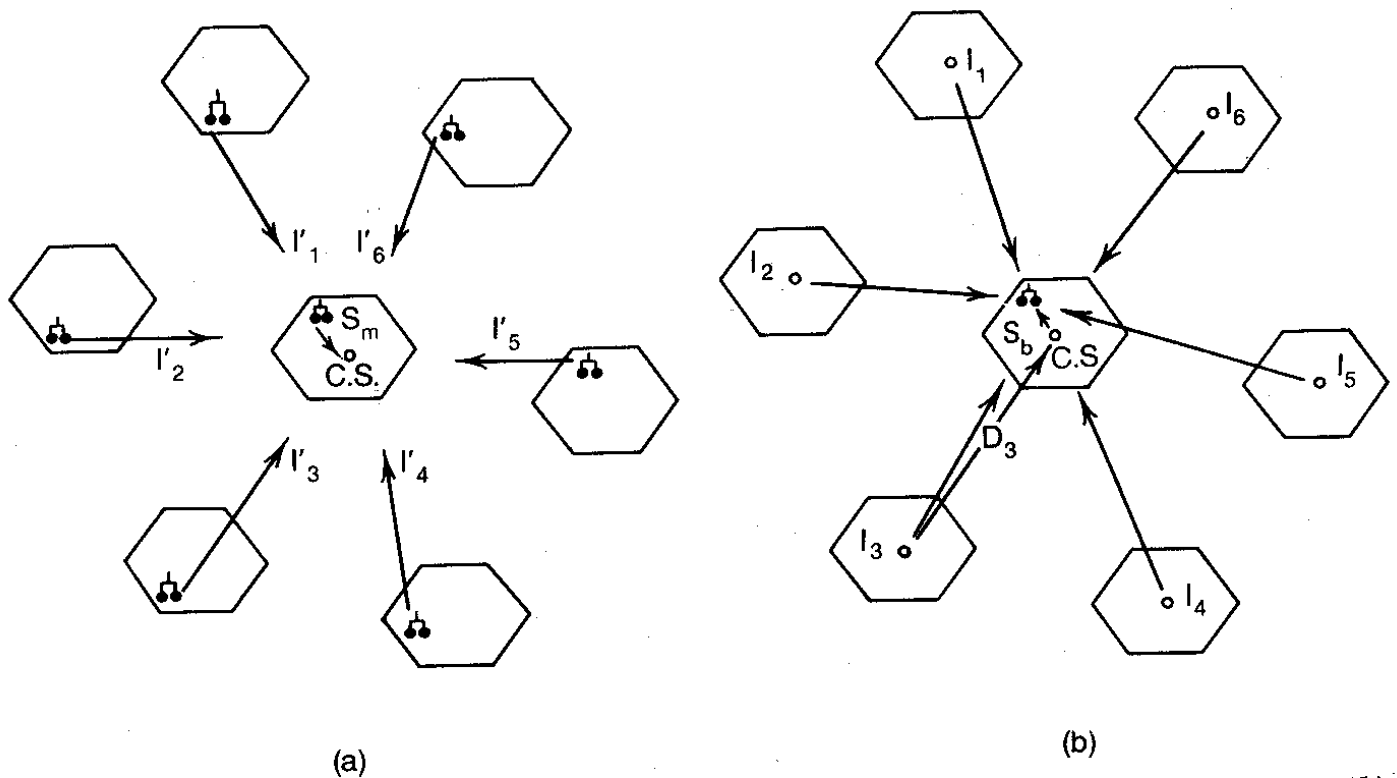


Figure 2.5 Cochannel interference from six interferers. (a) Receiving at the cell site; (b) receiving at the mobile unit.

1.4 Cell splitting

Cell splitting is a process which a larger cell is subdivided into smaller cell on some conditions like congestion, heavy traffic at that particular cell. Cell splitting is done in when traffic density is high the frequency channels F_i in each cell C_i is not able to handle enough mobile calls then the original cell can be split into smaller cells. The radius of new cell is half of the original cell radius as shown in figure 1.8. There are two ways of cell splitting, in figure 1.8(a) the original cell site is not used but in the figure 2.7 (b) original cell is used.

$$\text{New cell radius} = \frac{\text{old cell radius}}{2}$$

$$\text{New cell area} = \frac{\text{old cell radius}}{4}$$

Assume that maximum traffic load capacity of new cell is same as the old cell

$$\frac{\text{New Traffic Load}}{\text{Unit area}} = 4 \frac{\text{Traffic Load}}{\text{Unit area}}$$

There are two kinds of cell-splitting techniques:

1. Permanent splitting: This type of cell splitting is planned prior to the traffic load on the basis of the analytical data. Permanent cell splitting is done on the basis of number of channels, assigned frequencies, transmitted power and the traffic load of the original cell. The actual service of the original cell is stop for some time. This is done usually at midnight on a weekend.

2. Dynamic splitting: This scheme is based on using the allocated spectrum efficiency in real time. The algorithm for dynamically splitting cell sites is a tedious job, as we cannot afford to have one single cell unused during cell splitting at heavy traffic hours.

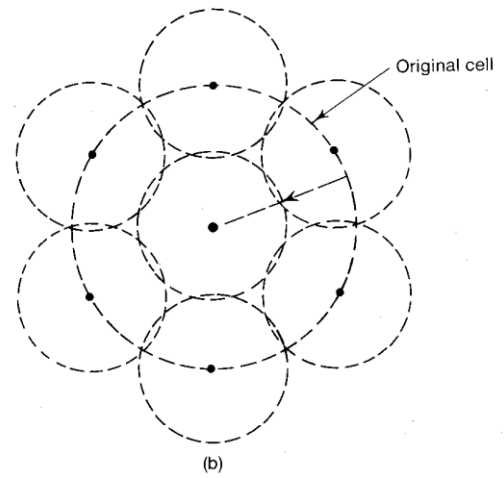
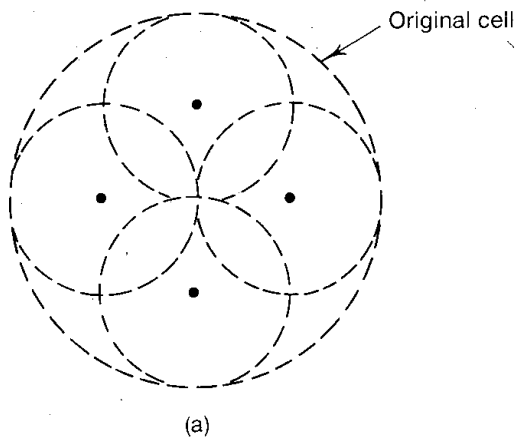


Figure 2.7 Cell splitting.

Cell Sectoring

The co-channel interference in cellular systems may be decreased by replacing a single Omni directional antenna at the base station by several directional antennas each radiating within a specified sector. By using directional antennas, a given cell will receive interference and transmit with only a fraction of the available co-channel cells. The technique for decreasing co-channel interference and thus increasing system performance by using directional antennas is called sectoring. The factor by which the co-channel interference is reduced depends on the amount of sectoring used. A cell is normally partitioned into three 120° sectors or six 60° sectors as shown in below figure.

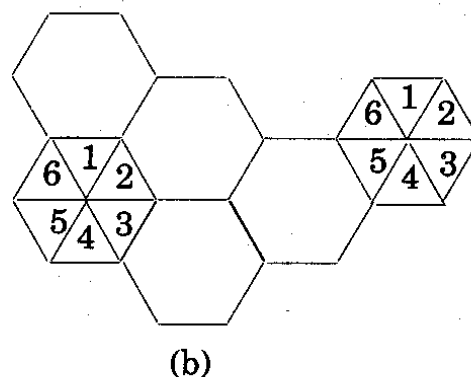
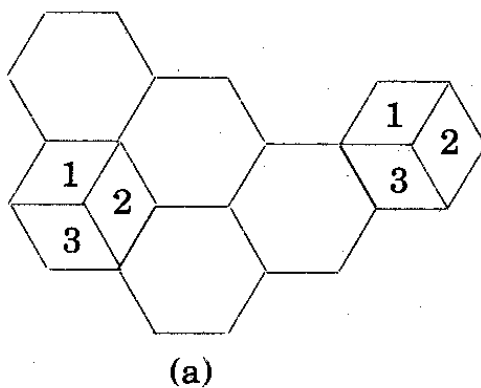


Fig: (a) 120° sectoring (b) 60° sectoring